YOUR SAFE ENERGY

LITHIUMVALLEY

SC100A50P-06W 100.3kWh/50kW

SC114A50P-06W 114.6kWh/50kW

A+ Grade Lithium Iron Phosphate Cells

15-year Design Lifespan

Version:V1.0

Features and Advantages

Safety Technology

- A+ grade lithium iron phosphate
- The system is configured with a multi-level fuse protection
- Built-in aerosol fire extinguishing systems at the package and rack levels

Easy to Install and Use

- Highly integrated for ease of transportation, installation and deployment
- Modular design with flexible system capacity configuration

More Reliable

- The system is rigorously designed and tested to operate stably under various conditions
- Equipped with protections against overcharge over-discharge, and over-temperature to prevent accidents

Energy Management System

- Emergency power supply
- Peak shaving
- Demand mitigation
- Self-consumption
- Micro-grids
- Price arbitrage

System Technical Parameters

Model	SC100A50P-06W	SC114A50P-06W
Battery Data		
Nominal Energy	100.3kWh	114.6kWh
Nominal Voltage	358.4V	409.6V
Battery Model	LV-IESS-RH14.336Aa-HBox-7	LV-IESS-RH14.336Aa-HBox-8
Battery Chemistry	Lithium Iron Phosphate(LiFePO4)	
Cycle Life	≥8000 cycles(@25°C, 0.5P/0.5P)	
Cell Specification	3.2V 280Ah	
Battery Pack Energy	14.336kWh	
Number Of Racks	1	
String Configuration	1P112S	1P128S
Operating Voltage	313.6V - 408.8V	358.4V - 467.2V
Rated Charging Power	0.5P	
Rated Discharging Power	0.5P	
Safety		
Fire Extinguishing System	Aerosol fire extinguishing	
Inverter Data		
Rated Power	50kW	
Please refer to the inverter section for more details.		
General		
Dimension(WxDxH)	1050 x 1180 x 2250mm	
Weight	Approximately 1200kg	Approximately 1250kg
IP Rating	IP55	
Cooling Method	Smart air cooling	
Operating Temperature	-20°C - +50°C	
Storage Temperature	-20°C - +45°C	
Relative Humidity	15% - 90%(Non-condensing)	
Operating Altitude	≤3000m	
Communication Interfaces	LAN/CAN/RS485	
Compliance	IEC 62619, IEC/EN 61000, UN 38.3, MSDS	

* The actual product may have slight differences from some promotional videos or pictures. Please refer to the actual product. Unless otherwise specified, the data on this page are all from our company's laboratory. The data may have errors due to changes in the objective environment.
* The specifications are subject to change without prior notice.

Hybrid Inverter

Three Phase | High Voltage

50kW

Inverter Features

- 4 Integrated MPPTs with string current capacity of up to 20A.
- Maximum charge/discharge current of up to 70A+70A on two independently controlled battery ports.
- Supports peak shaving control in both "self-use" and "generator" mode.
- Backup port supports 1.6 times overload for short durations.
- Generator connectivity with multiple input methods and automatic generator On/Of control.
- Allows parallel operation of up to 6 units for both on-grid and of-grid applications.
- Support to work as grid-tied inverter without battery connection during on-grid.
- Compatible with a wide voltage range to accommodate various batteries.

Inverter Technical Parameters Part 1

Model	S6-EH3P50K-H
Input DC (PV side)	
Recommended max PV array size	100 kW
Max. usable PV input power	96 kW
Max. input voltage	1000 V
Rated voltage	600 V
Start-up voltage	180 V
Mppt voltage range	150-850 V
Max. input current	4*40 A
Max. Short circuit current	4*60 A
Mppt number/max. Input strings number	4/8
Battery	
Battery type	Li-ion
Battery voltage range	150-800 V
Max. Charge / discharge power	55 kW
Max. Charge / discharge current	70 A*2(1)
No. Of battery inputs	2
Max. charge / discharge power of each input	35 kW
Communication	CAN/RS485
Output AC (Grid side)	
Rated output power	50 kW
Max. apparent output power	50 kVA
Rated grid voltage	3/N/PE, 220 V / 380 V ; 3/N/PE, 230 V / 400 V
Rated grid frequency	50 Hz / 60 Hz
Rated grid output current	76 A / 72.2 A
Max. output current	76 A / 72.2 A
Power factor	>0.99 (0.8 leading - 0.8 lagging)
THDI	< 3%
Input AC (Grid side)	
Max. AC passthrough current	152 A /144.4 A
Rated input voltage	3/N/PE, 220 V / 380 V ; 3/N/PE, 230 V / 400 V
Rated input frequency	50 Hz / 60 Hz
Input Generator	
Max. input power	50 kW
Rated input current	76 A / 72.2A
Rated input voltage	3/N/PE, 220 V / 380 V ; 3/N/PE, 230 V / 400 V

Inverter Technical Parameters Part 2

Model	S6-EH3P50K-H
Rated input frequency	50 Hz / 60 Hz
Output AC (Back-up)	
Rated output power	30 kW
Max. apparent output power	1.6 times of rated power, 2 s
Back-up switch time	< 10 ms
Rated output voltage	3/N/PE, 220 V / 380 V ; 3/N/PE, 230 V / 400 V
Rated frequency	50 Hz / 60 Hz
Rated output current	45.6 A / 43.3 A
THDV (@linear load)	< 2%
Eficiency	
Max. Eficiency	97.8%
EU Eficiency	97.4%
BAT charged by PV max. Eficiency	98.5%
BAT charged/discharged to AC max. Eficiency	97.5%
Protection	
Anti-islanding protection	Yes
Output over current protection	Yes
Short circuit protection	Yes
Integrated DC switch	Yes
DC reverse-polarity protection	Yes
Surge protection	DC Type II / AC Type II
Integrated AFCI 2.0	Optional
General Data	
Dimensions (W x H x D)	530 x 880 x 290 mm
Weight	73 kg
Topology	Transformerless
Self-consumption (night)	<35 W
Operating ambient temperature range	-25°C - +60°C
Relative humidity	0-95%
Ingress protection	IP66
Cooling concept	Intelligent redundant fan-cooling
Max. Operation altitude	4000 m

Inverter Technical Parameters Part 3

Model	S6-EH3P50K-H
Grid connection standard	G99, VDE-AR-N 4105/VDE V 0124, EN 50549-1/EN 50549-10, VDE 0126/UTE C 15/VFR:2019, NTS 631/RD 1699/RD 244/UNE 206006/UNE 206007-1, CEI 0-21, C10/11, NRS 097-2-1, TOR, EIFS 2018.2, IEC 62116, IEC 61727, IEC 60068, IEC 61683, EN 50530, MEA, PEA,PORTARIA Nº 140, DE 21 DE MARCO DE 2022
Safety/EMC standard	EC/EN 62109-1/-2, IEC/EN 61000-6-2/-4, EN 55011
Features	
PV connection	MC4 Quick connection plug
Battery connection	Terminal connector
AC connection	Terminal Block
Display	7.0" LCD display & Bluetooth + APP
Communication	CAN, RS485, Ethernet, Optional: Wi-Fi, Cellular, LAN

(1) Supporting parallel 140A input.

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